IDLE Shell 3.11.3

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```
Python 3.11.3 (tags/v3.11.3:f3909b8, Apr  4 2023, 23:49:59)
Type "help", "copyright", "credits" or "license()" for more
>>>
=== RESTART: C:/Users/Student/AppData/Local/Programs/Python/
Solution found!
Number of moves: 2
(1, 2, 3)
(4, 5, 6)
(7, 0, 8)

(1, 2, 3)
(4, 5, 6)
(7, 8, 0)


>>>
=== RESTART: C:/Users/Student/AppData/Local/Programs/Python/P
[1, 0, 0, 0, 0, 0, 0, 0]
[0, 0, 0, 0, 1, 0, 0, 0]
[0, 0, 0, 0, 0, 0, 0, 1]
[0, 0, 0, 0, 0, 1, 0, 0]
[0, 0, 1, 0, 0, 0, 0, 0]
[0, 0, 0, 0, 0, 0, 1, 0]
[0, 1, 0, 0, 0, 0, 0, 0]
[0, 0, 0, 1, 0, 0, 0, 0]
>>>
=== RESTART: C:/Users/Student/AppData/Local/Programs/Python/Py
0 0
4 0
0 3
4 3
1 3
3 0
1 0
3 3
0 1
4 2
Target reached!
>>>
=== RESTART: C:/Users/Student/AppData/Local/Programs/Python/Pyth
Solution:
```

```
>>>
=== RESTART: C:/Users/Student/AppData/Local/
0 0
4 0
0 3
4 3
1 3
3 0
1 0
3 3
0 1
4 2
Target reached!
>>>
=== RESTART: C:/Users/Student/AppData/Local/Pr
Solution:
SEND = 9567
MORE = 1085
MONEY = 10652
>>>
=== RESTART: C:/Users/Student/AppData/Local/Pro
Solution found:
(3, 3, 0)
(3, 1, 1)
(3, 2, 0)
(3, 0, 1)
(3, 1, 0)
(1, 1, 1)
(2, 2, 0)
(0, 2, 1)
(0, 3, 0)
(0, 1, 1)
(1, 1, 0)
(0, 0, 1)
>>>
```

5th.py - C:/Users/Student/AppData/Local/Programs/Python/Python311/5th.py (3.11.3)

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```
from collections import deque

def solve():
    start = (3, 3, 0)
    goal = (0, 0, 1)

    queue = deque([(start, [])])
    visited = {start}

    while queue:
        state, path = queue.popleft()

        if state == goal:
            print("Solution found:")
            for s in path + [state]:
                print(s)
            return

        m, c, boat = state

        moves = [(1,0), (2,0), (0,1), (0,2), (1,1)]

        for x, y in moves:
            if boat == 0:
                new = (m-x, c-y, 1)
            else:
                new = (m+x, c+y, 0)

            nm, nc, nb = new

            if 0 <= nm <= 3 and 0 <= nc <= 3:
                if (nm == 0 or nm >= nc) and ((3-nm) == 0 or (3-nm) >= (3-nc)):
                    if new not in visited:
                        visited.add(new)
                        queue.append((new, path + [state]))

solve()
```

Ln: 38 Col: 0

Ln: 25 Col: 0 Ln: 54 Col: 57

3rd.py - C:/Users/Student/AppData/Local/Programs/Python/Python311/3rd.py (3.11.3)

4th.py - C:/Users/Student/AppData/Local/Programs/Python/Python311/4th.py (3.11.3)

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```
from itertools import permutations

word1 = "SEND"
word2 = "MORE"
result = "MONEY"

letters = set(word1 + word2 + result)

for p in permutations(range(10), len(letters)):
    d = dict(zip(letters, p))

    if d['S'] == 0 or d['M'] == 0:
        continue

    send = 1000*d['S'] + 100*d['E'] + 10*d['N'] + d['D']
    more = 1000*d['M'] + 100*d['O'] + 10*d['R'] + d['E']
    money = 10000*d['M'] + 1000*d['O'] + 100*d['N'] + 10*d['E'] + d['Y']

    if send + more == money:
        print("Solution:")
        print("SEND =", send)
        print("MORE =", more)
        print("MONEY =", money)
        break
```

Ln: 25 Col: 0

3rd.py - C:/Users/Student/AppData/Local/Programs/Python/Python311/3rd.py (3.11.3)

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```
from collections import deque

def water_jug(a, b, target):
    queue = deque([(0, 0)])
    visited = set([(0, 0)])

    while queue:
        x, y = queue.popleft()
        print(x, y)

        if x == target or y == target:
            print("Target reached!")
            return

        states = [
            (a, y),                 # Fill jug A
            (x, b),                 # Fill jug B
            (0, y),                 # Empty jug A
            (x, 0),                 # Empty jug B
            (x - min(x, b-y), y + min(x, b-y)),   # A -> B
            (x + min(y, a-x), y - min(y, a-x))    # B -> A
        ]

        for state in states:
            if state not in visited:
                visited.add(state)
                queue.append(state)

# Jug A = 4 litres
# Jug B = 3 litres
# Target = 2 litres

water_jug(4, 3, 2)
```

Ln: 34

1ST.py - C:/Users/Student/AppData/Local/Programs/Python/Python311/1ST.py (3.11.3)

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```
    2: [1, 5],
    3: [0, 4, 6],
    4: [1, 3, 5, 7],
    5: [2, 4, 8],
    6: [3, 7],
    7: [4, 6, 8],
    8: [5, 7]
}

queue = deque([(start, [])])
visited = {start}

while queue:
    state, path = queue.popleft()

    if state == goal:
        print("Solution found!")
        print("Number of moves:", len(path))

        for step in path:
            for i in range(0, 9, 3):
                print(step[i:i+3])
            print()
        break

    blank = state.index(0)

    for position in moves[blank]:
        new_state = list(state)

        # Swap blank with adjacent tile
        new_state[blank], new_state[position] = \
            new_state[position], new_state[blank]

        new_state = tuple(new_state)

        if new_state not in visited:
            visited.add(new_state)
            queue.append((new_state, path + [new_state]))
```